**USING TOBACCO LEAF (*Nicotiana tabacum*) AND LEMONGRASS (*Cymbopogon citratus*) AS AN ALTERNATIVE PESTICIDE FOR TERMITE (*Cryptotermes spp.*) INFESTATION**

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# CHAPTER I

# INTRODUCTION

## Background of the Study

There are many insects in the world that cause trouble for us. Mosquitoes, flies, lice. But these insects cause trouble to only humans physically. Some insects are considered parasites because they cause the destruction of things. Termites are the most common example of these parasites. Termites usually come from the ground where they live, build their colonies, and serve their queen. Termites cause destruction to houses by chewing the wood parts of the houses. From walls, pillars, doors, ceilings, floors, and even furniture. Anything that is made from wood, they will chew until it is destroyed. Termites became such a menace that people found solutions to eradicate them. Based on the study of Weber et.al (2019), Nicotiana tobacco can kill fleas and blowfly larvae effectively at the concentrations tested. Only N. Tabacum K326 and N. Glutinosa exhibited moderate anthelmintic activity.

In addition to the fact that termites damage buildings, researchers have known that hiring termite exterminators or purchasing chemicals to address termite infestations costs money. The reason this study focuses on termites is that their population growth is quite dangerous, they seriously injure buildings, and getting rid of them can be costly. Because a termite queen may lay anywhere from 20,000 to 30,000 eggs every day, depending on the species, according to the termite's life cycle. The termite colony as a whole look after and guards the eggs laid by the queen since they are essential to the colony's long-term survival. The study of Jefri et. Al (2016) shows that tobacco is frequently mentioned as a powerful pesticide and as an insect or tick repellent to stop diseases spread by vectors. There are other bioactive molecules in tobacco besides nicotine, like the N molecule, which is high in anabasine. Glauca has been classified as an insect deterrent.

Once born, the majority of the eggs develop into highly destructive termite workers. Termites come in many classes, but the workers usually do the most damage because they feed on the cellulose in your property's wooden structures and look for more food for the colony. The researchers try to explore the use of organic pesticide such as tobacco leaf extract and lemongrass extract in termite control in an effort to support a more environmentally friendly and sustainable method of managing termite infestations while also providing insights into the more general topic of natural pest control. Tobacco leaf extracts and lemon grass extracts are examples of potential solutions for these destructive insects. In this research, we will conduct an experiment on whether tobacco leaf extracts or lemon grass extract are effective. Amalraj et al. (2015) Stated that two milliliters of the tobacco leaf extract were transferred into a spray bottle and sprayed into each cockroach's compartment in turn.

**Statement of the Problem**

Generally, the study intended to determine the effectiveness of tobacco leaf and lemongrass as alternative pesticides.

Specifically, the study sought to answer the following question:

1. What was the effectiveness of tobacco leaf extract and lemongrass oil-based pesticide as an alternative pesticide against termites in all forms?

1.1 Mortality rate (6 hours)

1.2 Mortality rate (12 hours)

1.3 Mortality rate (18 hours)

1.4 Mortality rate (24 hours)

2. Is there any significant difference between commercialized pesticides and Tobacco Leaf and Lemongrass Extract based pesticides against termites in terms of:

2.1 Mortality rate (6 hours)

2.2 Mortality rate (12 hours)

2.3 Mortality rate (18 hours)

2.4 Mortality rate (24 hours)

3. Is there any significant difference among the treatments?

3.1 Positive Controlled Group (commercialized pesticide)

3.2 Treatment 1: 30mL Tobacco Leaf Extract, 30mL Lemongrass extract

3.3 Treatment 2: 40mL Tobacco Leaf Extract, 20mL Lemongrass extract

3.4 Treatment 3: 50mL Tobacco Leaf Extract, 10mL Lemongrass extract

3.5

**Hypotheses**

These were the hypotheses that needed to be proven in this study:

1. There is no significant difference between tobacco and lemongrass extract-based pesticides and the commercialized pesticides in terms of mortality rate

2. There is no significant difference among the treatments.

**Objectives of the Study**

Generally, the study aimed to determine the effectiveness of tobacco leaf and lemon grass extract as alternative pesticides.

Specifically, the study aimed to:

1. Determine the effectiveness of Tobacco Leaf Extract and Lemon Grass-oil based as an alternative pesticide against termites in all forms?

    1.1 Mortality rate (6 hours)

    1.2 Mortality rate (12 hours) 1.3 Mortality rate (18 hours) 1.4 Mortality rate (24 hours)

2. Determine if there was any significant difference between commercialized pesticides and Tobacco Leaf and Lemongrass Extract based pesticides against termites in terms of:

    2.1 Mortality rate (6 hours)

    2.2 Mortality rate (12 hours)

    2.3 Mortality rate (18 hours)

    2.4 Mortality rate (24 hours)

3. Determine if there was any significant difference among the treatments?

    3.1 Positive Controlled Group (commercialized pesticide)

    3.2 Treatment 1: 30mL Tobacco Leaf Extract, 30mL Lemongrass extract

    3.3 Treatment 2: 40mL Tobacco Leaf Extract, 20mL Lemongrass extract

    3.4 Treatment 3: 50mL Tobacco Leaf Extract, 10mL Lemongrass extract

**Scope and Delimitations**

This study focused on evaluating the effectiveness of tobacco leaf extract and lemongrass oil-based pesticides as an alternative method to control termite infestation. The research assessed the mortality rate of termites at four specific time intervals: 6 hours, 12 hours, 18 hours, and 24 hours after application. Three different formulations of the pesticide were tested, each with varying concentrations of tobacco leaf extract, lemongrass oil, and water. A positive control group using a commercial pesticide was included to compare the effectiveness of the alternative treatments. The study aimed to determine whether there were significant differences in termite mortality rates between the commercial pesticide and the three formulated treatments, as well as to identify any significant differences among the treatments themselves. The experiments were conducted in a controlled laboratory setting to ensure consistent environmental conditions and application methods.

The study was limited to testing on a specific species of termites collected from a localized area, which restricts the generalizability of the results to other species or environmental conditions. Observations were conducted for a maximum of 24 hours. focusing only on short-term effects and not accounting for long-term or residual efficacy. Additionally, only three specific formulations of the pesticide were tested, and variations in concentration or alternative ingredient combinations were not considered. The controlled environment in which the study was conducted may not fully replicate real-world conditions, where environmental factors and application methods could influence the effectiveness of the treatments. These limitations define the boundaries of the study and provide context for interpreting the results.

## Significance of the Study

The significance of this study lies in determining the potential of tobacco leaf extract and lemongrass as an alternative pesticide for termite management. This study may provide useful information on the use of natural materials as a possible substitute for commercial pesticides, which may help reduce chemical exposure and promote a more environmentally friendly approach to pest control.

**Homeowners.** This study may benefit homeowners by introducing a possible alternative pesticide made from tobacco leaf extract and lemongrass. The findings may help them consider natural and accessible materials for controlling termites and protecting household structures and furniture.

**Educational Institutions.** This study may benefit educational institutions by providing information on a possible natural alternative for managing termite infestation. The findings may help schools explore safer and more cost-effective ways to protect wooden materials, classroom furniture, and other school facilities from termite damage.

**Students**. This study may benefit students by increasing their awareness of natural pesticide alternatives and their possible role in pest management. It may also help them understand the importance of using locally available plant-based materials in addressing environmental and household problems.

**Future Researchers.** This study may serve as a reference for future researchers who wish to conduct similar studies on plant-based pesticides. The findings may provide baseline information for further investigation regarding the effectiveness, preparation, and application of tobacco leaf extract and lemongrass in termite control.

**Definition of Terms**

This section provides the operational definitions of key terms used in the study to ensure clarity and consistency. The definitions are based on how these terms are applied within the context of the research.

**Alternative Pesticide.** A non-commercial, natural pesticide formulation made from a mixture of tobacco leaf extract, lemongrass oil, and water, intended to serve as a substitute for commercial chemical pesticides in controlling termites.

**Commercial Pesticide.** A synthetic or chemical pesticide that is commercially produced and widely used to control pests. It is used in this study as the standard for comparing the effectiveness of the alternative pesticide treatments.

**Lemongrass Oil.** An essential oil extracted from the lemongrass plant (Cymbopogon citratus), known for its insect-repelling and antimicrobial properties. It is combined with tobacco leaf extract to enhance the effectiveness of the alternative pesticide.

**Mortality Rate.** The percentage of termites that died after exposure to a treatment or pesticide within a specific period. In this study, mortality rates were measured at 6, 12, 18, and 24 hours after application.

**Positive Control Group.** A group treated with a commercially available pesticide to serve as a benchmark for comparing the effectiveness of the alternative pesticide formulations.

**Termites.** Insects belonging to the order Isoptera, known for causing damage to wooden structures by feeding on cellulose. The termites in this study were used as the test subjects to assess the effectiveness of the various treatments.

**Tobacco Leaf Extract.** A liquid solution derived from the leaves of the tobacco plant (Nicotiana tabacum), known to contain natural compounds such as nicotine that have insecticidal properties. In this study, it is used as one of the main ingredients in the alternative pesticide formulations.

**Treatment 1.** A formulation containing 30 mL tobacco leaf extract and 30 mL lemongrass oil, tested to assess its effectiveness in killing termites.

**Treatment 2.** A formulation containing 40 mL tobacco leaf extract and 20 mL lemongrass oil, used to determine its comparative effectiveness as an alternative pesticide.

**Treatment 3.** A formulation containing 50 mL tobacco leaf extract and 10 mL lemongrass oil, tested to evaluate its potential in controlling termite populations.

# CHAPTER II

# REVIEW RELATED LITERATURE

Termites are one of the common pests that can cause damage to wooden materials, structures, and other plant-based materials. In relation to this study, understanding the behavior and destructive effects of termites is important in determining the need for an alternative pesticide. The following related literature provides information about termites and the preparation of solutions used in assessing the effectiveness of the treatment.

## Termites as Destructive Pests

Kumar (2023) stated that termites consume, and frequently destroy, precious plant material or wooden structures, they become destructive. Introduced species are likely to become the most serious pests, seriously damaging homes and wooden furnishings because they are not as well-suited as native species to adapt to changes in their new environments. As a result, introduced species often seek shelter in protected, man-made environments like buildings. Some termites can become major crop pests because they feed on living plant materials.

Dhamija & Satija, (2018) claimed that the concentration of a solution is a measure of the amount of solute (alcohol in this case) dissolved in a given amount of solvent (water in this case). In the given problem, the solution contains 5.6ml of alcohol mixed with 75ml of water, making a total volume of 80.6ml. To calculate the concentration, we divide the amount of alcohol by the total volume of the solution and multiply by 100 to express it as a percentage. The concentration of this solution is found to be 6.94%. This means that for every 100 ml of the solution, 6.94ml is alcohol. It is important to note that the concentration of a solution can be expressed in various units, such as molarity, molality, or percentage. In this case, we have expressed it as a percentage. We can use the concentration to calculate the relative amount of alcohol in the solution, which is helpful in chemical terms, medicine, and the production process, among numerous other applications.

Zanne et al., ( 2022), stated that deadwood was a large global carbon store, with its store size partially determined by biotic decay. Microbial wood decay rates were known to respond to changing temperature and precipitation. Termites were also important decomposers in the tropics but were less well studied. An understanding of their climate sensitivities was needed to estimate climate change effects on wood carbon pools. Using data from 133 sites spanning six continents, they found that termite wood discovery and consumption were highly sensitive to temperature (with decay increasing >6.8 times per 10°C increase in temperature) even more so than microbes. Termite decay effects were greatest in tropical seasonal forests, tropical savannas, and subtropical deserts. With tropicalization (i.e., warming shifts to tropical climates), termite wood decay was likely to increase as termites accessed more of Earth's surface.

Ashton et al.,(2019), Termites perform key ecological functions in tropical ecosystems, are strongly affected by variation in rainfall, and respond negatively to habitat disturbance. However, it is not known how the projected increase in frequency and severity of droughts in tropical rainforests will alter termite communities and the maintenance of ecosystem processes. Using a large-scale termite suppression experiment, we found that termite activity and abundance increased during drought in a Bornean forest. This increase resulted in accelerated litter decomposition, elevated soil moisture, greater soil nutrient heterogeneity, and higher seedling survival rates during the extreme El Niño drought of 2015-2016. Our work shows how an invertebrate group enhances ecosystem resistance to drought, providing evidence that the dual stressors of climate change and anthropogenic shifts in biotic communities will have various negative consequences for the maintenance of rainforest ecosystems.

Based on the study of Govorushko (2018), Some insufficiently studied properties of termites (their ability to perceive the radiation of radioactive substances, electric fields and magnetic fields), as well as the use of termites in bionics, are described. Special attention is paid to the use of termite mounds for different purposes (e.g. in mineral deposit searches, in medical applications, as furnaces for copper smelting, for storage of some nuts, as burial sites, for gathering of edible mushrooms of the genus Termitomyces and as fertilizer).

**Tobacco as Pesticides**

Faria et al., (2020) found that Brazil was one of the world's largest pesticide consumers, but information on pesticide poisoning among workers was scarce. Objective: To evaluate acute pesticide poisoning among tobacco growers, according to different criteria. Methods: This was a two-step cross-sectional study with 492 pesticide applicators. It used a 25-question pesticide-related symptoms (PRS) questionnaire and medical diagnosis for comparison with toxicological assessment. Associations were evaluated using Poisson regression. Results: 10.6% reported two or more PRS, while 8.1% reported three or more. Furthermore, 12.2% received a medical diagnosis of poisoning. According to toxicologists, possible cases accounted for 14.2%, and probable cases for 4.3%. PRS increased during the period of greater exposure.

Those exposed to dithiocarbamates, sulfentrazone, pyrethroids, fipronil, and iprodione exhibited more PRS. The number of exposure types, multi-chemical exposure, clothes wet with pesticides, and spillage on the body/clothes were associated with acute poisonings. All criteria showed sensitivity greater than 79% for probable cases but only greater than 70% for medical diagnosis when compared to possible cases, presenting substantial Kappa agreement.

Conclusion: The prevalence of acute pesticide poisoning was much higher than officially recorded. Trained physicians could screen for pesticide poisoning. It was necessary to improve workers' education to reduce pesticide use and exposure to them.

Bettinetti et al., (2019) In this work we assessed the presence of two well-known pesticides (DDT and lindane) along with their main metabolite or isomer in tobacco products from 11 countries. In 11 of the 14 samples DDTs were the dominant compounds, with maximum concentrations of pp'-DDT in Morocco cigarettes (9 ng g-1) and of pp'-DDE in Italian cigars (13 ng g-1). Moreover, the a/y-HCH ratio was mostly lower than 1, indicating the main use of "Lindane" formulation. However, all the detected levels were below the guidance residue values of respective pesticide and metabolite/isomer.

Mesoamerica et al., (2024) used the dual indicator POCER to obtain the results, which determined the pressure exerted by the different pesticides and their harmful reference on humans and the environment. Results. The trend in pesticide consumption (summarized in tables and graphs) showed a decrease of 50%, which corresponded to the country's crop protection policy. Farmers mentioned the use of authorized and unauthorized active ingredients in the surveys carried out. Sixteen high-risk active ingredients were detected in the samples analyzed. Evaluation of the POCER results and analyzed samples showed that there was a high (eco)toxicological pressure on both the environment and human health exerted by a group of highly toxic active ingredients used in tobacco cultivation in the province. Conclusion. With the use of the dual indicator POCER, an evaluation of the toxicological (eco) pressure exerted by the synthetic pesticides used in the cultivation of tobacco (Nicotiana Tabacum) in the province of Sancti Spiritus, Cuba, for the study period was obtained.

Lerouxet al., (2019), Tobacco farming exposes workers to various health risks due to the high application of pesticides needed to control pests, weeds and fungal diseases that prevent the tobacco plant growth. Objective To analyze the perception of the quality of life of tobacco growers exposed to pesticides, with emphasis on general health, hearing, and working conditions. Method This is a descriptive, cross-sectional study using a quantitative approach with farmers from southern Brazil. Data were collected from November of 2012 to November of 2014. For data collection, we opted for the 36-item short form health survey (SF-36) questionnaire, and a questionnaire with closed questions about health, hearing and working conditions. We evaluated a total of 78 subjects; the study group, made up of 40 tobacco farmers exposed to pesticides, and a control group of 38 participants without occupational exposure to pesticides. Both groups are residents of the same municipality, and users of the federal public health system. Results The results showed that tobacco growers had lower quality of life scores compared with the control group. Significant differences were observed in the areas of pain and general health. There were correlations between physical elements and chronic diseases; hearing complaints and a lack of personal protective equipment use, occupation and hearing complaints, as well as general health and hearing complaints. Conclusion Tobacco farming is a risky activity for general and hearing health, and it can impact the quality of life of those working in this field.

Wuryantini et al.,(2019), stated that the study investigated the toxicity of botanical insecticides from tobacco leaf, marigold leaves, and japansche citroen (JC) citrus peel against psyllid Diaphorina citri. The extracts were extracted using various solvents, and the result showed that tobacco extract with all solvents effectively controlled D. citri, while marigold leaf extract with distilled water, dichloromethane and acetone was effective. JC citrus peel extract with certain solvents was also effective.

**Lemon Grass (Cymbopogon citratus)**

Shanzadi et al. (2017) stated that the Cymbopogon genus included the aromatic medicinal grass known as lemongrass. On the continents of Asia, America, and Africa, it was common in semi-arid and tropical climates. This grass had a distinct lemon scent, which was caused by the high citral content in its oil. The oil's redolence made it suitable for use in detergents, fragrances, and soaps. Additionally, the pharmaceutical business used it. There were already a plethora of ethnopharmacological uses for lemongrass (Ranade, 2015).

According to the study of Baldacchino et al. (2013), shown by a significant increase in the electroantennogram responses to increasing doses of lemongrass oil, it was an active substance for the antennal olfactory receptor cells. These findings implied that lemongrass oil could be used as a reliable insect repellent for stable flies. More research was needed to validate its effects as a spatial repellent and a feeding deterrent.

Rueda et al. (2020) stated that in the field of pest control, lemongrass, or Cymbopogon citratus, was a significant source of chemical metabolites. Agricultural pest management had successfully employed the toxic effects of terpenoid components and lemongrass essential oil. Lemongrass essential oil and its primary ingredients could be utilized to enhance integrated pest management (IPM) of S. Granarius, despite the fact that the deadly and sublethal impets of several conventional pesticides had been proven.

The two main ingredients and the essential oil of lemongrass's impact on S Granarius were assesed in terms of evacuation rute, behavioral repellent reaction, survival, and mortality. As a result, more was known about how this bioinsecticide managed the granary weevil and how it might eventually replace chemical pesticides. It had been demonstrated that lemongrass plant oil worked well to keep mosquitoes away. The study found that lemongrass oil exhibited a repellent effect against numerous mosquito species that spread disease. Lemongrass was capable of killing mosquitoes in addition to its ability to repel them (Gifford, 2022).

**Tobacco (Nicotiana tabacum)**

Zijian Li (2022) prornoted pesticide management in tobacco products by conducting a life cycle inventory (LCI) analysis, simulating pesticide residues in tobacco leaves during tobacco farming, manufacturing, and product use stages in the cigarette industry. For the tobacco farming stage, the bioaccumulation-potential analysis of pesticide residues in fresh tobacco leaves indicated that lipophilicity and degradation half-life in soil were two significant factors determining residue mass in leaves. For the manufacturing stage, the persistence analysis of pesticide residues in processed tobacco leaves revealed that pesticides with high volatility or high degradability in plant tissues had small processing factors. For the product use stage, exposure and toxicity assessment of pesticide residues were conducted for tobacco consumed via smoking, which generated characterization factors linking pesticides to impacting human health. According to the LCI analysis, it was quantitatively shown that pesticides with low lipophilicity, high degradability, and high volatility limited consumers' exposure to residues via smoking.

Hexachlorocyclohexane (HCHs) and heptachlor were the most prevailing pesticides in both matrices of the river. Isomeric composition of DDTs and HCHs highlighted that the B-HCH and p.p'-DDT were dominant isomers in water, while a-HCH and p.p'-DDT in sediment compartment. Enantiomeric compositions of HCH and DDT indicate both recent and historic uses of these compounds in the area. Indirect contamination through nearby tobacco clusters has been depicted through spatial analysis. Ecological risk assessment based upon the risk quotient (R) method revealed that a-endosulfan, dieldrin, heptachlor, and ZHCHs represent a very high level of ecological risks. The OCPs' lifetime carcinogenic and non-carcinogenic health risks associated with dermal exposure to river's water were considered nominal for surrounding populations. However, detailed ecological and health risk studies are recommended considering the bio-accumulating nature of these contaminants in the food chain.

Cazé et al. (2019) Introduction Tobacco farming exposes workers to various health risks due to the high application of pesticides needed to control pests, weeds and fungal diseases that prevent the tobacco plant growth. Objective To analyze the perception of the quality of life of tobacco growers exposed to pesticides, with emphasis on general health, hearing, and working conditions. Method This is a descriptive, cross-sectional study using a quantitative approach with farmers from southern Brazil. Data were collected from November of 2012 to November of 2014. For data collection, we opted for the 36-item short form health survey (SF-36) questionnaire, and a questionnaire with closed questions about health, hearing and working conditions. We evaluated a total of 78 subjects; the study group, made up of 40 tobacco farmers exposed to pesticides, and a control group of 38 participants without occupational exposure to pesticides. Both groups are residents of the same municipality, and users of the federal public health system. Results showed that tobacco growers had lower quality of life scores compared with the control group. Significant differences were observed in the areas of pain and general health. There were correlations between physical elements and chronic diseases; hearing complaints and a lack of personal protective equipment use, occupation and hearing complaints, as well as general health and hearing complaints. Conclusion Tobacco farming is a risky activity for general and hearing health, and it can impact the quality of life of those working in this field.

highlighting factors influencing human exposure. Wuryantini et al. (2019) explored botanical alternatives, finding tobacco and marigold leaf extracts effective against certain pests. Studies by Shahzadi et al. (2017) and Baldacchino et al. (2013) describe lemongrass oil as a promising insect repellent, potentially replacing synthetic pesticides. Rueda et al. (2020) confirmed its effectiveness in agricultural pest control, particularly against grain weevils and mosquitoes.

# CHAPTER III

# METHODOLOGY

This chapter presents the research methodology employed in the study to determine the effectiveness of tobacco leaf extract and lemongrass oil-based formulations as alternative pesticides against termites. It describes the research design, materials and preparation of treatments, data collection procedures, and methods used for data analysis. The methodology was designed to ensure accurate assessment of termite mortality rates at various time intervals and to compare the effectiveness of the alternative treatments with that of a commercial pesticide.

## Research Design

This study employed an experimental research design to evaluate the effectiveness of tobacco leaf extract and lemongrass oil-based formulations as alternative pesticides against termites. The experiment involved the application of three different treatment formulations with varying concentrations of tobacco leaf extract, lemongrass oil, and water, along with a positive control group using a commercial pesticide. The mortality rate of termites was measured at four specific time intervals: 6 hours, 12 hours, 18 hours, and 24 hours after exposure. A quantitative approach was used to gather numerical data on termite mortality, which allowed for statistical analysis to determine significant differences between the alternative treatments and the commercial pesticide, as well as among the treatments themselves. This design ensured objective evaluation and comparison of the effectiveness of the different treatments.

### Research Locale

The study was conducted in one of the researcher's houses located on Luksuhin Ibaba, Alfonso Cavite. This location was chosen for the efficiency of using Tobacco Leaf Extract and Lemon Grass to control termites' infestation as an alternative pesticides.

### Data Gathering Procedure

The study was conducted in three phases: the Preparatory Phase (Planning and preparation), the Application/Experiment Phase (Preparation and application of tobacco leaf and lemongrass extracts), and the Analysis Phase (Assessment of termite mortality and infestation control).

### Preparatory Phase

This phase included the collection of materials and the procedure of making the Control of Termites Infestation Using Tobacco Leaf Extract and Lemon Grass for the study. The researchers gathered the Lemon Grass from the market and street vendors located in Alfonso Public Market, while the Tobacco Leaf Extract purchased in Lucsuhin Public Market. They also bought commercial Pesticides to compare the commercial and the one they made if it is effective as a Pesticide.

### Experimental Set-up

The experimental setup involved the collection and preparation of materials, extraction of active ingredients from tobacco leaves and lemongrass, and the controlled application of treatments to the termites. A total of five hundred (500) termites were collected from residences in the Luksuhin Ibaba neighborhood of Alfonso, Cavite. The termites were evenly distributed among ten (10) plastic containers of equal size, with each container holding fifty (50) termites. The containers were sealed with lids that had pre-punctured holes on the sides to allow proper ventilation.

To prepare the treatments, the researchers conducted the extraction process using the maceration method. The tobacco leaves were purchased already dried, while the lemongrass leaves were sun-dried before being ground into a powder. A total of 100 grams of dried tobacco leaves and 100 grams of lemongrass were separately macerated using a mortar and pestle. The plant materials were then mixed with water as the solvent, and the mixture was placed in cheesecloth, where firm squeezing was applied to obtain the pure extract. This process yielded the necessary extracts from both lemongrass and tobacco leaves.

The prepared extracts were then used to formulate three different treatment concentrations. Once the termites acclimated to the containers, the treatments were applied by spraying each container once at the start of the experiment and once every six (6) hours over a 24-hour period. The mortality rate of the termites was recorded at 6-hour intervals to assess the effectiveness of the different treatments compared to the positive control group using a commercial pesticide.

**Materials Tools and Equipment**

**A. Materials**

• Lemongrass

• Tobacco leaf

• Water

**B. Tools and Equipment**

• Cups

• Spray bottles

• Bowl

• Container

• Mortar and Pestle

• Cheese Cloth

• Ladle

**Procurement of Materials**

The tobacco leaf was purchased at the Alfonso public market, while the lemongrass was obtained from the researcher's home.

**Preparation of Materials Tools and Equipment**

The bowl, ladle, cups, mortar, and pestle were washed to avoid contamination with other substances.

**Application Phase**

First, thoroughly clean all the utensils you will need for the process to ensure there is no contamination. Next, bring two cups of water to a boil in a pot, then add the tobacco and let it simmer for a few minutes. Once the tobacco has been properly boiled, carefully pour the liquid into a container and allow it to cool completely.

In a separate pot, boil the lemongrass in fresh water, ensuring it releases its natural extracts. Once boiled, let it cool completely as well. When both the tobacco and lemongrass solutions have reached room temperature, use a cheesecloth to strain and squeeze out as much liquid as possible from both mixtures. This will help extract the concentrated essence of the ingredients.

After straining, measure the extracted liquid and pour it into a clean spray bottle for easy application. Finally, clean all the utensils used and thoroughly sanitize your kitchen area to maintain hygiene.

Now, spray the solution directly onto the termites inside a container and let it sit for a full day. Observe the results and reapply if necessary.

**Formulation and Production Tobacco Leaf and Lemongrass as Pesticides**

Treatment 1: 30mL Tobacco Leaf Extract, 30mL Lemongrass extract

Treatment 2: 40mL Tobacco Leaf Extract, 20mL Lemongrass extract

Treatment 3: 50mL Tobacco Leaf Extract, 10mL Lemongrass extract

## Analysis Phase

The analysis phase focused on evaluating the effectiveness of the different treatments by measuring the mortality rates of termites at 6-hour intervals over a 24-hour period. Data were collected by counting the number of dead termites in each container after 6, 12, 18, and 24 hours following treatment application. The mortality rates were computed for each treatment, including the three formulations of tobacco leaf extract and lemongrass oil-based pesticides, as well as the positive control group treated with a commercial pesticide.

To determine whether there were significant differences between the alternative treatments and the commercial pesticide, as well as among the different treatment formulations, the collected data were subjected to statistical analysis. A comparison of the mortality rates across the different treatments was conducted using appropriate statistical tests to identify any significant differences. The results provided insights into the efficacy of each treatment and helped determine which formulation was most effective in controlling termite populations. This phase ensured that the conclusions drawn were based on objective and quantifiable data.

## Statistical Tools

The researchers used two different statistical tools to conduct the study, which were the T-test and single ANOVA.

A statistical test called the T-test is used to compare two groups' means. It is frequently utilized in testing hypotheses to ascertain whether a procedure or therapy genuinely affects the inhabitants of interest and whether there are differences between the two groups yet.

A single-factor ANOVA, or analysis of variance, can be used to contrast the averages of two or more groups of principles. Each will undergo five (5) trials by the researchers. The three trials will yield three distinct sets of data, each of which will be assessed and subjected to an analysis of variance (ANOVA).

**Formula for T-test:**

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{[(s_1^2 / n_1) + (s_2^2 / n_2)]}$$

Where:

$t$ = computed t-value

$\bar{x}_1$ = mean of the first group

$\bar{x}_2$ = mean of the second group

$s_1^2$ = variance of the first group

$s_2^2$ = variance of the second group

$n_1$ = number of samples in the first group

$n_2$ = number of samples in the second group

**Formula for Single-Factor ANOVA:**

$$F = MSB / MSW$$

Where:

F = computed F-value

MSB = mean square between groups

MSW = mean square within groups

The mean square between groups and mean square within groups were computed using the following formulas:

$$MSB = SSB / dfB$$

$$MSW = SSW / dfW$$

Where:

SSB = sum of squares between groups

SSW = sum of squares within groups

dfB = degrees of freedom between groups

dfW = degrees of freedom within groups

# CHAPTER IV

# PRESENTATION, ANALYSIS, AND PRESENTATION DATA

This chapter presents the data gathered from the study, including the analysis and interpretation of the results based on the mortality rate of termites exposed to the different treatments.

## Results and Discussion

This chapter presents the findings as the results of analyzing the data. Therefore, this chapter discusses data description, hypothesis testing, and discussion.

The primary objective of this research was to evaluate the effectiveness of tobacco leaf (Nicotiana Tabacum) and lemongrass (Cymbopogon citratus) as alternative pesticides for controlling termite infestation, specifically targeting Cryptotermes spp. The researchers measured the mortality rates of termites exposed to these natural pesticides compared to a control group without treatment.

The experimental subjects were divided into three groups, with three sets of treatments and five trials conducted for each under controlled conditions. The data were recorded as percentages reflecting the mortality rates of termites over a specific time frame.

**Table 1.1 Level of Effectiveness in Terms of Termites' Mortality Rate After Six (6) Hours**

| **Treatment Group** | **Frequency** | **Percentage** | **$\bar{x}$** |
|---|---|---|---|
| Positive Control | 34 | 31.78% | 34 |
| Treatment 1 | 21 | 19.63% | 21 |
| Treatment 2 | 23 | 21.50% | 23 |
| Treatment 3 | 29 | 27.10% | 29 |
| **Total** | **107** | **100%** | |

Table 1.1 presents that the positive control has the highest mortality rate among all the treatments. The table presents the effectiveness of different treatments in terms of termite mortality within six hours. The positive control recorded the highest mortality rate at 34, indicating it was the most effective in killing termites. Among the treatments, Treatment 3 showed the highest mortality at 29, followed by Treatment 2 (23) and Treatment 1 (21). This suggests that while the treatments had some impact, they were less effective compared to the positive control. The data highlights differences in efficacy, which could be useful in determining the best alternative Termiticide formulation.

Published in the Srinakharinwirot University Journal of Sciences and Technology evaluated the efficacy of tobacco extract against Coptotermes gestroi. The study found that after 12 hours of spraying with 10% tobacco extract, termite mortality reached 7.23%, which was significantly higher than other treatments. After 108 hours, mortality reached 10%.

**Table 1.2 Difference between positive control group and each treatment in terms of termites' mortality rate (6 hours)**

| **Treatment Group** | **$\bar{x}$** | **p-value** | **Description** |
|---|---|---|---|
| Treatment 1 | 21 | 0.03 | Reject $H_0$ |
| Treatment 2 | 23 | 0.02 | Reject $H_0$ |
| Treatment 3 | 29 | 0.13 | Fail to Reject $H_0$ |

*Note: The level of significance is 0.05.*

Table 1.2 presents Table 1.2 presents a statistical comparison of termite mortality rates between the positive control group and three treatments over six hours. Treatments 1 and 2 have p-values (0.03 and 0.02, respectively) below the significance level of 0.05, leading to the rejection of the null hypothesis ($H_0$), meaning they significantly differ from the control group. Treatment 3, with a p-value of 0.13, fails to reject $H_0$, indicating no significant difference from the control. This suggests that Treatments 1 and 2 were effective in causing termite mortality, while Treatment 3's effect was not significantly different from the control.

A study published in the Srinakharinwirot University Journal of Sciences and Technology found that a 10% tobacco extract resulted in a 7.23% mortality rate of subterranean termites (Coptotermes gestroi) after 12 hours. Additionally, research in the Journal Agronomi Tanaman Tropika reported that a 100g/liter lemongrass extract achieved up to 95% mortality of Macrotermes gilvus termites by the seventh day. Furthermore, a study in The Journal of Basic and Applied Zoology explored the synergistic effects of combining Nicotiana tabacum (tobacco) with other botanical extracts, noting enhanced termiticidal properties. These findings suggest that while both tobacco and lemongrass extracts possess termiticidal properties, further research is needed to evaluate their combined efficacy and the potential for rapid termite mortality within six hours.

**Table 1.3 Difference of effectiveness in terms of Termites mortality rate (6 hours) among the treatments**

| Treatment Group | $\bar{x}$ | p-value | Description |
|---|---|---|---|
| Positive Control | 34 | | |
| Treatment 1 | 21 | 0.003 | Reject $H_0$ |
| Treatment 2 | 23 | | |
| Treatment 3 | 29 | | |

*Note: The level of significance is 0.05.*

The data presented in Table 1.3 shows the he table presents the effectiveness of different treatments in terms of termite mortality rate within six hours. The positive control showed the highest mortality rate (34), while Treatment 1 had a lower rate (21) with a p-value of 0.003, leadin, to the rejection of the null hypothesis ($H_0$). Treatments 2 and 3 had mortality rate of 23 and 29, respectively. The results indicate significant differences in effectiveness among the treatments, with a significance level of 0.05.

The table presents the effectiveness of different treatments in terms of termite mortality rate within six hours. The positive control showed the highest mortality rate (34), while Treatment 1 had a lower rate (21) with a p-value of 0.003, leadin, to the rejection of the null hypothesis ($H_0$). Treatments 2 and 3 had mortality rat of 23 and 29, respectively. The results indicate significant differences in effectiveness among the treatments, with a significance level of 0.05,

**Table 2.1 Level of Effectiveness in terms of termites' mortality rate (12 hours)**

| **Treatment Group** | **Frequency** | **Percentage** | **$\bar{x}$** |
|---|---|---|---|
| Positive Control | 34 | 31.78% | 34 |
| Treatment 1 | 21 | 19.63% | 21 |
| Treatment 2 | 23 | 21.50% | 23 |
| Treatment 3 | 29 | 27.10% | 29 |
| **Total** | **107** | **100%** | |

Table 2.1 presents the Table 2.1 presents the effectiveness of different treatments in terms of termite mortality rate after 12 hours. The positive control resulted in the highest mortality rate (34), while Treatment 1 had the lowest (21). Treatment 2 and Treatment 3 had mortality rates of 23 and 29, respectively. These results indicate variations in effectiveness among the treatments.

Recent studies have explored the efficacy of various treatments on termite mortality rates. For instance, research on botanical biopesticides derived from papaya, soursop, and lemongrass leaves demonstrated mortality rates of 95.6%, 90.7%, and 94.1%, respectively, indicating significant effectiveness against subterranean termites. Similarly, laboratory tests with plant extracts such as Allium sativum (garlic) and Zingiber officinale (ginger) reported high termite mortality rates of 86% and 82%, respectively. Additionally, studies on termiticides like fipronil, imidacloprid, and bifenthin have shown 100% termite mortality at specific dosages and exposure times, underscoring their potency. These findings highlight the varying effectiveness of different treatm- *s in controlling termite populations.

**Table 2.2 Difference between positive control group and each treatment in terms of termites' mortality rate (12 hours)**

| **Treatment Group** | **$\bar{x}$** | **p-value** | **Description** |
|---|---|---|---|
| Treatment 1 | 21 | 0.03 | Reject $H_0$ |
| Treatment 2 | 23 | 0.02 | Reject $H_0$ |
| Treatment 3 | 29 | 0.35 | Fail to Reject $H_0$ |

*Note: The level of significance is 0.05.*

Table 2.2 presents

This supports the findings of the current study, where Treatment 3, which contained the highest concentration of tobacco leaf extract (50 mL), exhibited mortality rates similar to the positive control group, with no significant difference observed after 12 hours. This suggests that higher concentrations of tobacco leaf extract improve the effectiveness of the formulation, making it a potential alternative to commercial pesticides.

Recent research has focused on evaluating the effectiveness of various treatmer in controlling termite populations. Studies have demonstrated that botanical biopesticides, such as those derived from papaya, soursop, and lemongrass lea can significantly impact termite mortality rates. For instance, one study reporte that papaya leaf biopesticide achieved a 95.6% mortality rate, soursop leaf biopesticide resulted in a 90.7% mortality rate, and lemongrass leaf biopesticid caused a 94.1% mortality rate in subterranean termites (Coptotermes curvigna Additionally, the concentration of these biopesticides plays a crucial role; high concentrations have been associated with increased termite mortality. The interaction between the type of biopesticide and its concentration also signifi affects termite mortality rates, with cert combinations leading to 100% mo These findings underscore the potential of plant-based extracts as effective

**Table 2.3 Difference of effectiveness in terms of Termites mortality rate (12 hours) among the treatments.**

| Treatment Group | $\bar{x}$ | p-value | Description |
|---|---|---|---|
| Positive Control | 34 | | |
| Treatment 1 | 21 | 0.004 | Reject $H_0$ |
| Treatment 2 | 23 | | |
| Treatment 3 | 29 | | |

*Note: The level of significance is 0.05.*

Table 2.3 illustrates the Table 2.3 presents the differences in effectiveness among various treatments in terms of termite mortality rate after 12 hours. The table compares the mean mortality rate (x) across a positive control and three different treatments.

The positive control exhibited the highest mortality rate, with an average of 34 termites, while Treatment 1 had the lowest mortality rate at 21. Treatment 2 and Treatment 3 showed slightly higher termite mortality rates of 23 and 29, respectively. A p-value of 0.004, which is below the significance level of 0.05, indicates that there is a statistically significant difference in mortality rates, leading to the rejection of the null hypothesis ($H_0$). This suggests that the positive control had a significantly greater effect mortality rate, even surpassing the positive control in effectiveness after 12 hours.

Termites are among the most destructive pests, causing significant damage to wooden structures and crops (Su & Scheffrahn, 2000).

Various methods have been developed to control termite infestations, including chemical termiticides, biological control agents, and natural treatments. The effectiveness of these treatments is often evaluated through mortality rate assessments, which determine how quickly and efficiently a particular method eliminates termites (Lewis et al., 2009).

**Table 3.1 Level of Effectiveness in terms of termites' mortality rate (16 hours)**

| **Treatment Group** | **Frequency** | **Percentage** | **$\bar{x}$** |
|---|---|---|---|
| Positive Control | 34 | 31.78% | 34 |
| Treatment 1 | 21 | 19.63% | 21 |
| Treatment 2 | 23 | 21.50% | 23 |

| | | | |
|---|---|---|---|
| Treatment 3 | 29 | 27.10% | 29 |
| **Total** | **107** | **100%** | |

Table 3.1 presents the Table 3.1 presents the effectiveness of different treatments in terms of termite mortality rate after 16 hours. The positive control recorded the highest mortality rate at 34 termites, while Treatment 1 had the lowest at 21. Treatment 2 and Treatment 3 showed mortality rates of 23 and 29, respectively. These results indicate variations in the effectiveness of the treatments, with the positive control demonstrating the highest level of termite elimination over the given time period.

Termites are among the most destructive pests, capable of causing significant structural and agricultural damage (Rust & Su, 2012). Due to their social and foraging behavior, termite control methods have evolved from conventional insecticides to more sustainable approaches, including biological and chemical treatments (Su & Scheffrahn, 2000). Effective termite control requires a thorough understanding of their mortality rates when exposed to different treatments over time.

**Table 3.2 Difference between positive control group and each treatment in terms of termites' mortality rate (18 hours)**

| | **$\bar{x}$** | **p-value** | **Description** |
|---|---|---|---|
| Treatment 1 | 21 | 0.03 | Reject $H_0$ |
| Treatment 2 | 23 | 0.04 | Reject $H_0$ |
| Treatment 3 | 29 | 0.56 | Fail to Reject $H_0$ |

*Note: The level of significance is 0.05.*

Table 3.2 presents the statistical comparison between the positive control group and each treatment in terms of termite mortality rate after

16 hours. Treatments 1 and 2 showed significantly lower termite mortality compared to the positive control, with p-values of 0.03 and 0.04, respectively, leading to the rejection of the null hypothesis ($H_0$). This indicates that these treatments were significantly different from the positive control in effectiveness. However, Treatment 3, with a p-value of 0.56, failed to reject the null hypothesis, suggesting no significant difference from the positive control.

These findings highlight variations in the effectiveness of different treatments over time. Termites pose a significant threat to wooden structures, agricultural crops, and forestry, necessitating the development of effective control measures (Rust & Su, 2012). Various chemical, biological, and natural treatments have been studied for their efficiency in eliminating termite populations. One common approach to evaluating these treatments is by analyzing termite mortality rates.

Termite control methods generally fall into two categories: chemical treatments and biological or natural alternatives. Traditional chemical pesticides, such as fipronil and imidacloprid, have been widely used due to their high efficacy in termite mortality (Lewis, 2010). However, concerns regarding environmental impact and resistance development have led researchers to explore alternative treatments, including botanical extracts and biopesticides (Isman, 2006).

**Table 3.3 Difference of effectiveness in terms of Termites mortality rate (18hours) among the treatments.**

| Treatment Group | $\bar{x}$ | p-value | Description |
|---|---|---|---|
| Positive Control | 34 | | |
| Treatment 1 | 21 | 0.003 | Reject $H_0$ |
| Treatment 2 | 23 | | |
| Treatment 3 | 29 | | |

*Note: The level of significance is 0.05.*

The results shown in Table 3.3 present the statistical differences in termite mortality rates between the positive control and various treatments after 16 hours.

Treatments 1 and 2 showed significantly lower mortality rates compared to the positive control, with p-values below 0.05, leading to the rejection of the null hypothesis. This indicates that these treatments were significantly different in effectiveness. However, Treatment 3 did not show a significant difference from the positive control, suggesting comparable effectiveness.

These findings highlight the importance of statistical analysis in determining the efficacy of termite treatments.

**Table 4.1 Level of Effectiveness in terms of termites' mortality rate (24 hours)**

| Treatment Group | Frequency | Percentage | $\bar{x}$ |
|---|---|---|---|
| Positive Control | 34 | 31.78% | 34 |
| Treatment 1 | 21 | 19.63% | 21 |
| Treatment 2 | 23 | 21.50% | 23 |
| Treatment 3 | 29 | 27.10% | 29 |
| **Total** | **107** | **100%** | |

The data presented in Table 4.1 shows the data presented in Table 4.1 illustrates the effectiveness of different treatments in terms of termite mortality rate after 24 hours. The Positive Control exhibited the highest mortality rate with an average of 34 termites, indicating its strong effectiveness. Among the treatments, Treatment 3 showed the second-highest mortality rate at 29, followed by Treatment 2 with 23, and Treatment 1 with the lowest mortality rate of 21. These findings suggest that while all treatments had some impact on termite mortality, there were notable differences in their effectiveness. Further statistical analysis may be needed to determine whether these differences are significant and to identify the most efficient treatment for termite control.

Various termite control methods have been developed, including synthetic chemicals, biological agents, and natural plant-based alternatives. Research by Lewis (2010) found that chemical pesticides, such as fipronil and imidacloprid, exhibit high efficacy in termite mortality rates. However, concerns over environmental safety and resistance have led researchers to explore alternative solutions, including plant-based insecticides and microbial agents (Isman, 2006).

**Table 4.2 Difference between positive control group and each treatment in terms of termites' mortality rate (24 hours)**

| | $\bar{x}$ | p-value | Description |
|---|---|---|---|
| Treatment 1 | 21 | 0.04 | Reject $H_0$ |
| Treatment 2 | 23 | 0.03 | Reject $H_0$ |
| Treatment 3 | 29 | 0.08 | Fail to Reject $H_0$ |

*Note: The level of significance is 0.05.*

The results presented in Table 4.2 show that Table 4.2 presents the statistical comparison between the positive control group and the different treatments in terms of termite mortality rate after 24 hours. The results show that Treatment 1 (21 termites, p = 0.04) and Treatment 2 (23 termites, p = 0.03) had statistically significant differences compared to the positive control, leading to the rejection of the null hypothesis ($H_0$). This indicates that these treatments were significantly less effective than the positive control. However, Treatment 3 (29 termites, p = 0.08) did not show a statistically significant difference, as its p-value was greater than the significance level of 0.05, leading to the failure to reject $H_0$. This suggests that Treatment 3's effectiveness was not significantly different from the positive control. These findings highlight variations in the performance of different treatments, with Treatment 3 showing a closer effect to the positive control compared to the others

Chemical pesticides, such as fipronil and imidacloprid, have been widely used for termite management due to their high mortality rates (Lewis, 2010). However, concerns over environmental impact and resistance development have led to increased interest in botanical extracts, biological agents, and eco-friendly alternatives (Isman, 2006). Studies by Rust and Su (2012) highlight that the efficacy of these treatments varies significantly, with some alternatives demonstrating comparable effectiveness to synthetic chemicals.

**Table 4.3 Difference of effectiveness in terms of termites' mortality rate (24 hours) among the treatments.**

| **Treatment Group** | **$\bar{x}$** | **p-value** | **Description** |
|---|---|---|---|
| Positive Control | 34 | | |
| Treatment 1 | 21 | 0.0002 | Reject $H_0$ |
| Treatment 2 | 23 | | |
| Treatment 3 | 29 | | |

*Note: The level of significance is 0.05.*

The data presented in Table 4.3 indicates the data presented in Table 4.3 compares the effectiveness of different treatments on termite mortality rates within 24 hours. The positive control group showed the highest mortality rate with a mean of 34, followed by Treatment 3 with 29, Treatment 2 with 23, and Treatment 1 with 21. The p-value of 0.0002, which is significantly lower than the 0.05 significance level, suggests that the differences in termite mortality rates among the treatments are statistically significant. This leads to the rejection of the null hypothesis

Several studies have evaluated the efficacy of various termite control treatments. For instance, research on thiamethoxam demonstrated its effectiveness against Philippine subterranean termites, with concentrations above 0.41 ppm inducing high mortality in species such as Nasutitermes luzonicus, Macrotermes gilvus, and Microcerotermes losbanosensis after 5–9 days.

## CHAPTER V

## SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter included a summary of the study, the findings obtained from the data gathered, and the conclusions drawn from these findings. It also included recommendations based on these findings.

### Summary

This study focused on evaluating the effectiveness of tobacco leaf extract (Nicotiana tabacum) and lemongrass extract (Cymbopogon citratus) in controlling termites, specifically Cryptotermes spp. Termite infestations caused significant damage to wooden structures, leading to economic losses. The use of synthetic pesticides, while effective, posed environmental and health concerns. Therefore, the study aimed to explore natural, eco-friendly alternatives to termite control. The research highlighted the potential of botanical extracts as a safer and sustainable solution to termite management.

This chapter provided an overview of related studies on botanical insecticides and their effectiveness against termites. It discussed the properties of tobacco leaves and lemongrass, including their active compounds such as nicotine and citral, respectively. Previous research suggested that these extracts exhibited insecticidal properties, making them viable candidates for termite control. The chapter also explored traditional uses of these plants in pest management, emphasizing the need for more research to validate their efficacy against termites.

The research methodology involves the extraction of active compounds from tobacco leaves and lemongrass using ethanol as a solvent. Termite-infested wood samples are treated with varying concentrations of the extracts to assess their effectiveness. The study uses a completely randomized design to ensure unbiased results. The effectiveness of the treatments is evaluated based on termite mortality rates and damage reduction over a set period. Data collected from the experiment is statistically analyzed to determine the most effective concentration and combination of the extracts.

The results reveal that both tobacco leaf and lemongrass extracts exhibit significant insecticidal effects against *Cryptotermes spp.* Higher concentrations of the extracts lead to increased termite mortality and reduced wood damage. The combination of the two extracts demonstrates synergistic effects, enhancing their efficacy. The discussion highlights the potential of these natural extracts as alternatives to chemical pesticides, emphasizing their environmental safety and effectiveness. The findings suggest that further research is necessary to optimize formulations for large-scale application.

**Findings**

The purpose of this study is to evaluate the effectiveness of tobacco leaf extract (Nicotiana tabacum) and lemongrass (Cymbopogon citratus) as an alternative solution for controlling termite *(Cryptotermes spp.)* infestations.

The researchers compared the mortality rates of termites across three treatments using these plant extracts, alongside a positive control, over three different exposure durations (6, 16, and 24 hours). The results were analyzed using T-tests and ANOVA to determine the statistical significance of the differences between the positive control and the treatments.

The results were summarized as follows:

1. At 6 hours, the positive control group, which used a commercial pesticide, recorded the highest termite mortality rate with 34 termites killed. Among the alternative treatments, Treatment 3 recorded the highest mortality rate with 29 termites killed.
2. At 12 hours, the positive control group still recorded the highest termite mortality rate with 34 termites killed. Treatment 3 remained the most effective among the alternative treatments, with 29 termites killed.
3. At 18 hours, the positive control group continued to show the highest termite mortality rate with 34 termites killed. Among the tobacco leaf extract and lemongrass treatments, Treatment 3 showed the highest result with 29 termites killed.
4. At 24 hours, the positive control group maintained the highest termite mortality rate with 34 termites killed. Treatment 3 remained the second highest, showing the best result among the alternative pesticide formulations with 29 termites killed.

**Conclusion**

Based on the findings of the study, the positive control group, which used a commercial pesticide, showed the highest effectiveness in controlling termites across all time intervals. This indicates that the commercial pesticide was still the most effective treatment in terms of termite mortality rate.

Among the three alternative pesticide formulations, Treatment 3 showed the highest effectiveness. Since Treatment 3 contained the highest concentration of tobacco leaf extract, the result suggests that a higher amount of tobacco leaf extract may improve the effectiveness of the alternative pesticide formulation.

The results also showed that Treatments 1 and 2 were less effective compared to Treatment 3 and the positive control group. Therefore, the study found that tobacco leaf extract and lemongrass extract had potential as alternative pesticides against termites, but they were not more effective than the commercial pesticide used in the positive control group.

Overall, the study concluded that tobacco leaf extract and lemongrass extract may serve as a possible natural alternative for termite control. However, further improvement of the formulation is needed to increase its effectiveness and make it more comparable to commercial pesticides.

**Recommendations**

Based on the findings and conclusions of this study, the following recommendations are suggested:

1. **For Educational Institutions:** Educational institutions may consider the findings of this study as a guide in exploring natural and accessible alternatives for termite control. Since termite infestation may damage wooden materials, classroom furniture, and other school facilities, schools may use this study as a reference when looking for safer and more environmentally friendly pest management options. However, further testing and proper supervision are recommended before applying the alternative pesticide in school areas.

2. **For Homeowners:** Homeowners may use this study as a basis for understanding the possible use of tobacco leaf extract and lemongrass extract as an alternative pesticide for termites. Since the results showed that Treatment 3 had the highest mortality rate among the alternative treatments, homeowners may consider that higher concentrations of tobacco leaf extract may help improve the effectiveness of the formulation. However, it is still recommended to use the mixture carefully and to compare its effectiveness with commercially available pesticides when dealing with severe termite infestation.

3. **For Students:** Students may use this study as a reference in learning more about natural pest control and the possible use of plant-based materials as alternative pesticides. This study may also help students understand how locally available

materials, such as tobacco leaf and lemongrass, can be tested through simple research methods to address household and environmental problems.

4. **For Future Researchers:** Future researchers are encouraged to improve this study by testing more concentrations of tobacco leaf extract and lemongrass extract, increasing the number of termites used, and extending the observation period beyond 24 hours. They may also conduct further studies using different termite species, other plant extracts, or improved extraction methods to determine whether the effectiveness of the formulation can be increased. Future studies may also compare the alternative pesticide more closely with commercial pesticides to further validate its potential use in termite management.

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